

Universal Short-Circuit and Open-Circuit Fault Protection and Detection for a Power Converter's Power Device

Buck F. Brown III , Graduate Student Member, IEEE, Zheyu Zhang, Senior Member, IEEE, Johan H. Enslin , Life Fellow, IEEE, and Cheng Wan, Senior Member, IEEE

Abstract—Short-circuit and open-circuit faults of a converter's power device often lead to catastrophic failure of the entire system if not detected and acted upon within a sub-microsecond time frame, particularly for emerging wide bandgap (WBG) power semiconductors. While much research has been done on the fast and accurate protection and detection of short-circuit faults, there has been less success corresponding to open-circuit faults. Common downfalls include protection and detection that are too application-specific, take longer than a couple of microseconds, and are not cost-efficient. This paper proposes a new open-circuit fault protection and detection system integrated with a pre-existing short-circuit system called desaturation protection. First, a literature review is conducted to confirm the necessity of the new protection and detection scheme. Second, the operation principle and design considerations of the newly proposed protection and detection circuitry are discussed. Third, a simulation and experiment are designed, conducted, and compared to showcase the new scheme's success, effectiveness, and adaptability in an actual inverter. The results of testing the proposed system illustrate the success and reliability of the new fault protection and detection scheme.

Index Terms—Converter protection, desaturation protection, open-circuit faults, reverse current faults.

I. INTRODUCTION

THE growth of electrical energy production throughout the past few decades is unmatched. As this rise in electrical energy production increases, the demand for reliable electrical components used in electrical energy production and conversion increases [1], [2]. In fact, the United States Department of Energy (DOE) expects 80% of electricity to flow through

power electronic converters by 2030 as the nation pushes toward ambitious renewable energy penetration [3]. Unfortunately, generic circuit breakers are no longer sufficient to protect power electronic converters, which often fail due to the sensitive switching components within them, such as emerging Silicon Carbide (SiC) MOSFETs or Gallium Nitride (GaN) HEMTs, being damaged and producing a fault, even leading to cascaded catastrophic failure of the entire converter [4], [5]. As a result, specialized fault protection and detection in converters is needed to prevent such calamities.

Many previous efforts have been made to solve the problem of fault protection and detection in converters. The two most common faults that occur are short-circuit (SC) faults and open-circuit (OC) faults. SC faults result in abnormally large currents. As these currents increase, the components' temperatures within the converter increase, leading to converter failure. This permanent damage occurs on a sub-microsecond scale, especially for WBG power semiconductors [6], [7]. As a result, SC faults within converters have been extensively studied. One solution, called desaturation (desat) protection, is widely applied and documented in [8].

OC faults also have drastic effects on components within converters. Most power electronic loads are inductive, so when an OC occurs, and there is no place for the current through the load to flow, there is a massive voltage spike across the opened power device due to high di/dt . The components within the converter cannot withstand such high voltages for extended periods, so the converter is damaged.

As with the SC fault, there have been many attempts to protect against and detect OC faults. However, unlike the SC fault, there lacks a universal, fast, and inexpensive OC fault protection and detection system.

In [9] and [10], an OC fault detection scheme for grid-connected inverters is proposed. The solution proposed in [9] consists of an in-depth, fault-type identification algorithm. Similarly, the solution proposed in [10] utilizes a novel sequential sorting algorithm. However, [9] focuses on fault detection in a wind energy conversion topology, and [10] focuses on fault detection in a grid-connected, neutral point clamped topology. Even though [9] and [10] pertain to grid-connected inverters, the solutions proposed in each are quite different because of their respective niche applications. There is a trend of application-specific solutions throughout the related literature. Each solution

Received 4 April 2024; revised 1 July 2024; accepted 6 August 2024. Date of publication 17 September 2024; date of current version 31 January 2025. Paper 2024-PEDCC-0367.R1, presented at the 2023 11th International Conference on Power Electronics and ECCE Asia, Jeju Island, South Korea, May 22–25, and approved for publication in the IEEE Transactions on Industry Applications by the Power Electronic Devices and Components Committee of the IEEE Industry Applications Society [DOI: 10.23919/ICPE2023-ECCEAsia54778.2023.10213883]. (Corresponding author: Buck F. Brown III.)

Buck F. Brown III, Johan H. Enslin, and Cheng Wan are with the Department of Electrical and Computer Engineering, Clemson University, North Charleston, SC 29405 USA (e-mail: buckb@g.clemson.edu; jenslin@clemson.edu; cwan@clemson.edu).

Zheyu Zhang is with the Electrical, Computer, and Systems Engineering, Rensselaer Polytechnic Institute, Troy, NY 12180 USA (e-mail: zhangz49@rpi.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIA.2024.3462910>.

Digital Object Identifier 10.1109/TIA.2024.3462910

TABLE I
COMPARISON OF STATE-OF-THE-ART SOLUTIONS

Reference	General Solution	Fast ($<5 \mu\text{s}$) Response Time	Low Price	Detection Capability
[9]	×	×	×	SC and OC
[10]	×	×	×	OC
[11]	×	×	×	OC
[12]	×	×	✓	OC
[13]	×	×	✓	SC and OC
[14]	✓	×	×	OC
[15]	✓	×	✓	OC
[16]	✓	×	✓	OC
[17]	✓	×	✓	OC
Proposed	✓	✓	✓	SC and OC

proposed in [9], [10], [11], [12], [13] is significantly different due to its application-specific nature.

The speed and cost of protection and detection are two other significant issues pertaining to protecting against and detecting OC faults in converters. In [14], a solution is proposed for OC fault detection in modular multilevel converters. This solution works by distributing fault diagnosis among various submodules. While this submodular approach provides the unquestionable benefit of reliable fault detection, it requires a minimum of 3.5 ms before protection and detection can occur. Newer data-driven and model-data-hybrid driven methods—[15] and [16], respectively—take advantage of the latest developments in neural networks to achieve OC fault detection. As a result, they are incredibly robust and adaptable. However, they take a minimum of several switching periods (over 100 μs) before they protect against and detect OC faults because of computational delays. Additionally, some solutions require multiple sensors, like those proposed in [9], [10], [11] and [14]. Because of their hardware-level implementation, these solutions can pinpoint where OC faults occur. Unfortunately, the required additional sensors drastically increase the cost of these solutions, making many of them infeasible for industry implementation.

One of the most complete solutions is given by [17]. The solution proposed in [17] pertains to OC fault detection and isolation for IGBTs in cascaded multilevel converters. This is accomplished through the combination of two sensors per converter phase leg and algorithmic processing of the measured data. This solution does not require a set number of individual modules in the converter to operate properly, and while it does pertain to cascaded multilevel converters, it is foreseeable that its approach could be modified for implementation in other converter topologies. However, the solution is only guaranteed to respond to OC faults in less than one fundamental cycle, which is not fast enough to prevent catastrophic failure of the converter under specific types of OC faults.

A summary of the capabilities of each solution proposed in [9], [10], [11], [12], [13], [14], [15], [16], [17] is given in Table I. Based on this table, there is no general, fast, and cost-efficient state-of-the-art solution capable of protecting against and detecting OC and SC faults. Therefore, it is essential to propose a new design for power devices capable of protecting against and detecting SC and OC faults within a few microseconds that

is also universal and inexpensive. This paper proposes such a solution.

The novelty of the proposed solution comes in the form of a power device level, sensor-less fault protection and detection scheme. Because of its hardware-level implementation, the solution proposed in this paper does not contain any computational algorithm delay and is faster than other solutions. Furthermore, it can be utilized in any converter topology, making it universal. Additionally, its complete lack of sensors makes it inexpensive. Finally, the flexibility of the scheme allows for it to have a relatively temperature-independent nature. Through its implementation, the most common types of faults in power devices can be protected and detected, preventing failure of the entire converter system.

This paper begins by describing the basic concept of the newly proposed fault protection and detection scheme. Then, the design methodology of the new scheme is detailed, and design considerations are given. Afterward, a simulation and experiment are designed, conducted, and compared to showcase the capabilities of the scheme. Finally, a cost analysis is performed to highlight the scheme's inexpensive nature.

II. DESIGN CONCEPT

This section begins by conducting an OC fault analysis to motivate the design of the new fault protection and detection scheme. Afterward, the new scheme built upon the foundation of desaturation protection is introduced. It should be noted that the subsequent analysis pertains to converters that are mid-operation and experience a fault while supplying some form of load current.

A. Open-Circuit Fault Analysis

There are two different classifications of converter OC faults. An OC fault can occur when current is flowing through the power device in the reverse direction, either from source to drain or through the freewheeling diode, or when current is flowing through the power device in the forward direction, from drain to source (using MOSFETs' terminology as an example) [18], [19], [20]. Fig. 1(a) and (b) show current flowing through a power device in the reverse and forward directions, respectively.

If current flows through the power device in the reverse direction, it can flow through either the channel or the freewheeling diode of the power device. Since there is nowhere for reverse current to flow if an OC fault occurs, a massive voltage spike dictated by high di/dt occurs in the circuit, as depicted by Fig. 1(a). As a result, the device and the surrounding converter will likely be permanently damaged.

If an OC fault occurs when current flows through the power device in the forward direction, the current through the load flows through the freewheeling diode of the complementary power device, as illustrated by Fig. 1(b). Therefore, there is no massive voltage spike across the faulty power device due to high di/dt , and catastrophic failure is avoided. However, having a dysfunctional power device is still a control issue, and the system cannot function properly without repair. Additionally, since there is no immediate, apparent damage to the circuit, it is

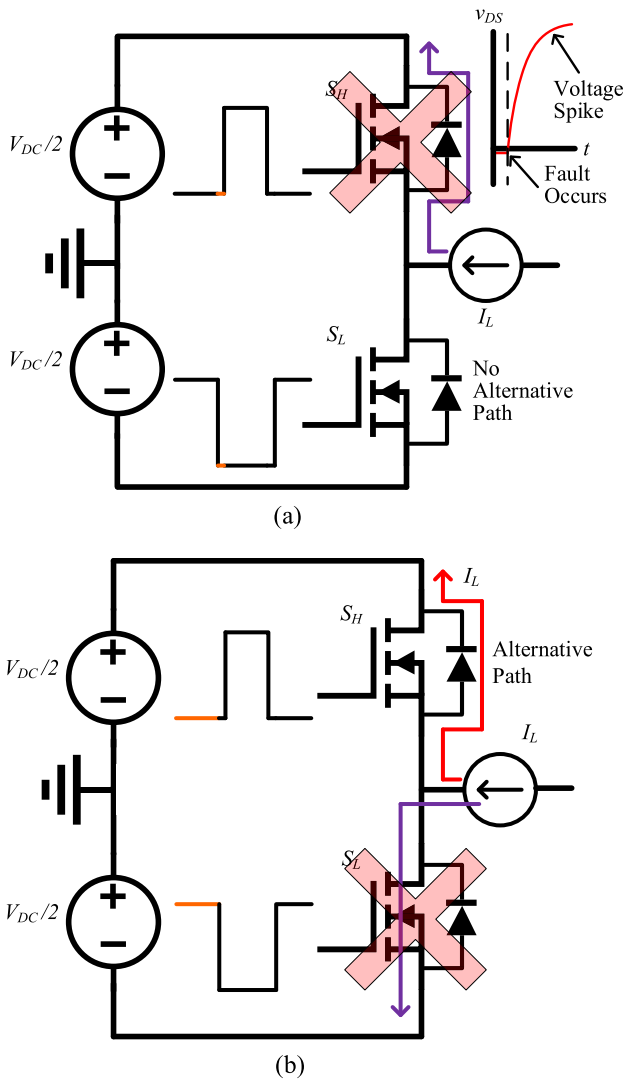


Fig. 1. Example of (a) reverse current open-circuit fault and (b) forward current open-circuit fault.

challenging to detect when forward current open-circuit faults occur. Hence, it is still essential to detect this type of open-circuit fault.

Clearly, reverse current OC faults cause more immediate, irreversible damage to the surrounding system than forward current OC faults. Hence, the probability that an OC fault leads to catastrophic failure of the entire system is approximately 0.5 in a two-level converter. It simply depends on the direction the current is flowing through the power device when the OC fault occurs. However, this analysis may not be as straightforward when examining multilevel converters. It is conceivable that in such a converter, regardless of the direction of current through the power device that experiences the OC fault, imminent failure of the converter system does not occur. This is because in certain scenarios, there may be additional paths through which the load current can flow, mitigating a voltage spike due to high di/dt . Despite this possibility, it is still vital that all OC faults are protected against and detected because even in the best-case

scenario, the converter system cannot function properly without repair.

A note is given here about the two most common types of power devices in converter circuitry—MOSFETs and IGBTs. MOSFETs contain an internal body diode as their freewheeling diode. Thus, when an open-circuit fault occurs, it is assumed that both the channel and the freewheeling diode are inoperative. However, IGBTs contain an external diode as their freewheeling diode. Thus, when an OC fault occurs, it is extremely unlikely that both the channel of the IGBT and its corresponding freewheeling diode are both inoperative. Despite the previous analysis focusing on MOSFETs, the analysis of IGBTs yields similar results. The worst-case scenario for IGBTs is when the freewheeling diode experiences an OC fault when current is flowing in the reverse direction, mimicking Fig. 1(a). Regardless of the type of power device used, the new protection and detection scheme needs to be capable of protecting against and detecting all OC faults.

B. Requirement of Fault Protection and Detection Scheme

The new fault scheme must protect against and detect reverse current OC faults, forward current OC faults, and SC faults. In each case, the proper sequence of protection and detection needs to be considered.

As discussed previously, if a reverse current OC fault occurs, a massive voltage spike can result due to high di/dt . Therefore, simply detecting the fault and turning off the power device does not protect the circuit. To protect the circuit from the voltage spike, the load current needs an alternative path to flow. After this current is redirected, the scheme must then detect the fault. Hence, the new scheme must first protect against and then detect reverse current OC faults. Unfortunately, none of the solutions given in [9], [10], [11], [12], [13], [14], [15], [16], [17] consider this; thus, the protection first, detection second approach given in this paper is another unique aspect of the proposed solution.

If the power device is turned off during a forward current OC fault or an SC fault, catastrophic damage from these faults can be avoided. Therefore, if these faults are detected quickly, and the faulty power device is subsequently turned off, they can be protected. Hence, the new scheme can first detect and then protect against forward-current OC and SC faults.

C. Fault Protection and Detection

A functional diagram of desaturation protection and the newly proposed OC fault protection and detection scheme are given in Fig. 2(a) and (b), respectively. By utilizing desaturation protection, a widely adopted technique shown in Fig. 2(a) designed to protect against and detect SC faults, the new scheme shown in Fig. 2(b) can protect against and detect both types of OC and SC faults. The new scheme has three main parts: a protection subsystem, a detection subsystem, and a bipolar comparator subsystem. The protection subsystem is achieved by a voltage clamper across the power device, and the detection subsystem detects the voltage across the power device. By using the bipolar comparator subsystem to determine when this detected voltage has passed some positive or negative voltage indicative of a fault, all types of faults are detected.

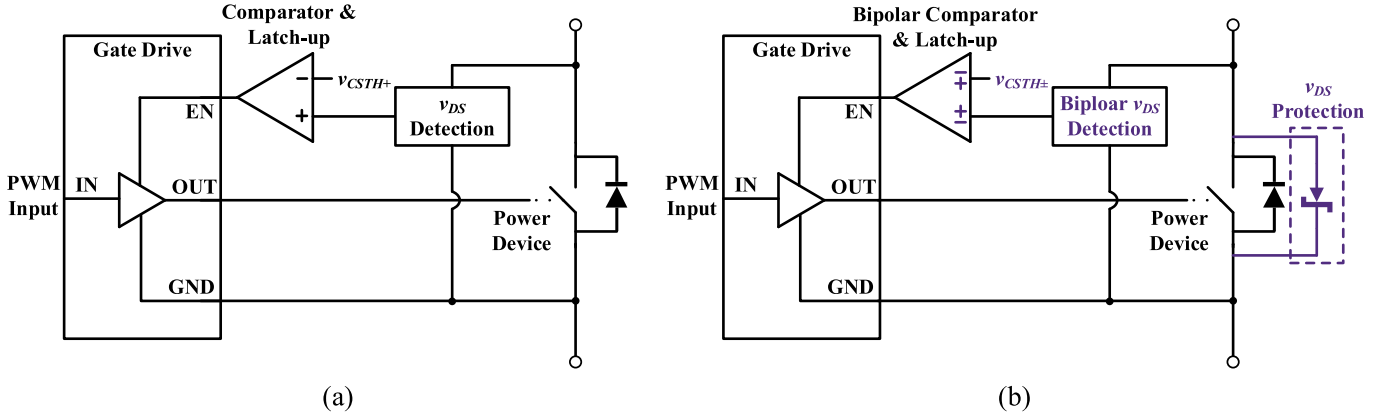


Fig. 2. Functional diagram of (a) desaturation protection and (b) newly proposed open-circuit fault protection and detection scheme.

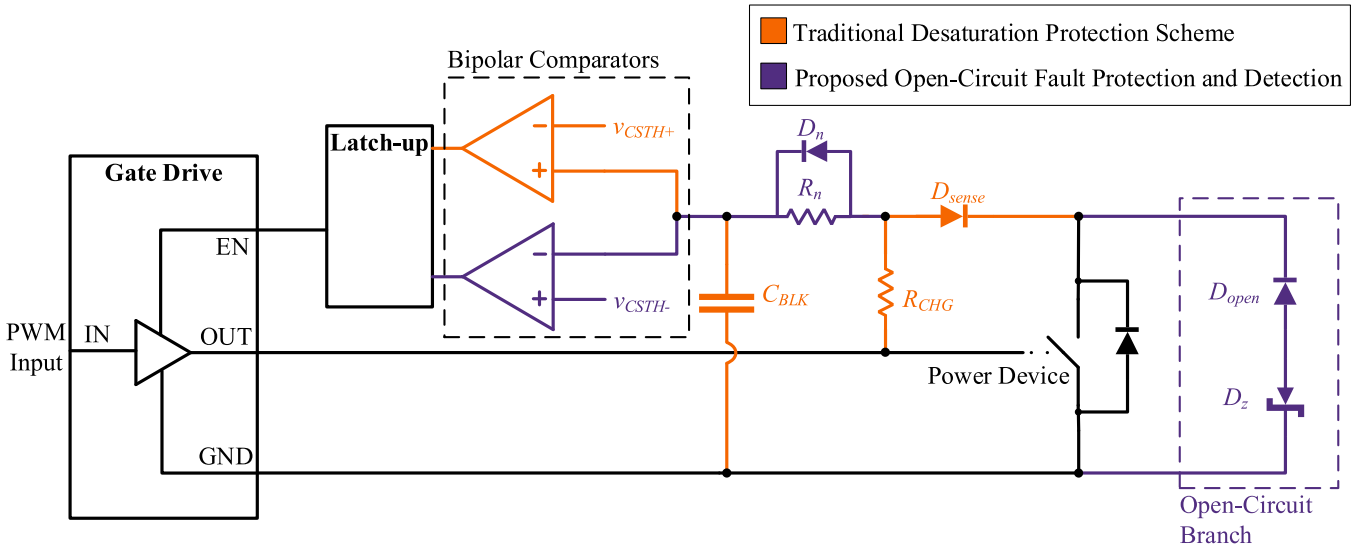


Fig. 3. Generic power device with protection and detection circuit and gate drive.

When a reverse current OC fault occurs, i.e., the fault demonstrated in Fig. 1(a), the new scheme first protects against and then detects the fault. Thus, the first function of the new scheme is to provide an alternative path consisting of a voltage clamping unit for current to flow only when a reverse current OC fault occurs. In doing so, the power device and the entire system are protected. Furthermore, the voltage clamping unit has a negative clamping voltage that is greater in magnitude than the normal negative on-state voltage of the power device when reverse current flows. Since the drain-to-source voltage of the power device is already being measured by desaturation protection, and providing the alternative path for current to flow produces a unique, negative voltage across the power device determined by the voltage clamper, another comparator can be introduced into the circuit to determine when this negative voltage occurs. Therefore, a reverse current OC fault is detected if the drain-to-source voltage exceeds some predefined negative threshold voltage.

When a forward current OC fault occurs, i.e., the situation depicted in Fig. 1(b), the drain-to-source voltage of the power device becomes the DC-bus voltage. Since this voltage exceeds a

predefined positive threshold voltage corresponding to this type of OC fault, the OC fault will be detected, and the power device will be turned off.

When SC faults occur, a huge spike in current is produced. This increase in current results in an increase in the drain-to-source voltage of the power device [8]. By monitoring the drain-to-source voltage, the desaturation protection circuitry detects when SC faults occur by comparing the drain-to-source voltage of the power device with a predefined positive threshold voltage. A fault is detected if the drain-to-source voltage exceeds the positive threshold voltage of desaturation protection. Subsequently, the power device is turned off, protecting the rest of the system from the abnormally large current produced by the SC fault.

III. DESIGN METHODOLOGY

The detailed proposed fault protection and detection system is illustrated in Fig. 3. This section discusses the newly proposed system. First, the SC detection and protection the circuitry

provides is mentioned, and then the novel OC protection and detection the circuitry offers is described. Afterward, practical design considerations are given.

A. Desaturation Protection Design

A standard desaturation protection circuit is the foundation of the circuit shown in Fig. 3. It consists of the following: a charging resistor, R_{CHG} ; a sensing diode, D_{sense} ; a blanking capacitor, C_{BLK} ; a positive comparator with a positive threshold voltage, v_{CSTH+} ; and a latch-up circuit. Design details are given in [8] and will not be repeated here.

B. Additional Open-Circuit Fault Protection

The standard desaturation protection circuit must be enhanced to protect against and detect reverse current OC faults. These enhancements include three addendums to the traditional desaturation protection circuitry. The first addendum is the insertion of a branch in parallel with the drain and source terminals of the power device consisting of a diode, D_{open} , in anti-series with a voltage clamper, D_z . The second addendum is the inclusion of a noise immunity resistor, R_n , and its corresponding bypass diode, D_n . The third addendum is the introduction of a bipolar comparator with a negative threshold voltage, v_{CSTH-} .

Adding the branch consisting of a diode in anti-series with a voltage clamper diminishes the catastrophic failure due to a reverse current OC fault and allows for fault detection. During regular operation, current does not flow through this branch because the voltage clamper blocks the current. When a reverse current OC fault occurs, the voltage clamper enters its breakdown region, allowing current to flow through the branch while also generating a distinguishable negative voltage for OC fault detection. As a result, the voltage across the branch is clamped to a specific value given by

$$v_{OC} = -v_{f_Dopen} - v_{z,b} \quad (1)$$

where v_{f_Dopen} is the forward voltage of the diode and $v_{z,b}$ is the breakdown voltage of the voltage clamper. The voltage $v_{z,b}$ is up to the user to decide. Whereas traditionally there is no place for the current flowing through the load to go when a reverse current OC fault occurs, it now has an alternative path.

The noise immunity resistor and corresponding bypass diode are essential to prevent the false triggering of the detection system due to switching ringing and interference [21]. They prevent the blanking capacitor from drastic fluctuations in the sensed voltage. Under a reverse current OC fault, the discharge current flows through the blanking capacitor and noise immunity resistor, as shown in Fig. 4(a). Under an SC fault, the discharge current flows through the charging resistor, bypass diode, and the blanking capacitor, as shown in Fig. 4(b). Hence, the desired blanking time is achieved in both scenarios, delaying detection by approximately a microsecond, which is determined by the maximum switching time of the device under investigation. However, it is important to note that the protection given by the new OC branch is ubiquitous.

Due to the nearly constant negative voltage across the newly implemented OC branch due to the voltage clamper, it is easy

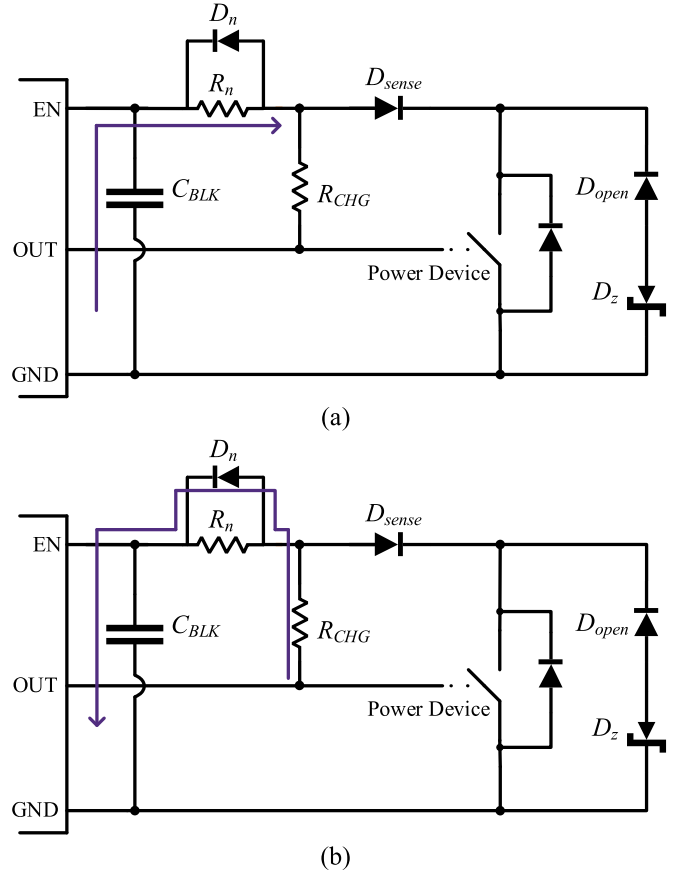


Fig. 4. Current discharge path under (a) reverse current open-circuit fault and (b) short-circuit fault.

to detect when a reverse current OC fault occurs. The negative comparator leverages the voltage across the blanking capacitor and compares it with a negative threshold voltage given by

$$v_{CSTH-} = -(v_{f_Dopen} + v_{z,b}) \cdot x \quad (2)$$

where x is a scaling factor between 80–100% to compensate for potential nonidealities and ensure detection of the OC fault. The outputs of the negative and positive comparators are tied together before the latch-up circuit. This is done practically through an OR gate. Therefore, regardless of the fault type, the latch-up circuit output indicates if a fault has occurred.

In summary, the proposed solution only consists of five new components: two diodes, one voltage clamper, one comparator, and one resistor. While the type and number of additional components may seem elementary, the way in which they are connected provides additional OC protection and detection functionality that is certainly not.

C. Design Considerations

A few considerations still need to be given before the circuit in Fig. 3 can be implemented. In particular, special design considerations need to be made for the charging resistor, the sensing diode, the blanking capacitor, the voltage clamper, and the standard diode contained in the OC branch. By adhering to the proceeding design considerations, the novel solution

proposed in this paper becomes easier to implement in practical environments.

The first component that needs to be reconsidered is the charging resistor, R_{CHG} . The resistor must have a power rating of

$$P_R = \frac{v_R^2}{R_{CHG}} \quad (3)$$

where

$$v_R = v_{GS} - (v_{DS} + v_{f_Dsense}) \quad (4)$$

Under a traditional desaturation protection scheme, v_R is always less than v_{GS} since both v_{DS} and v_{f_Dsense} are positive values. However, with the additional reverse current OC fault detection functionality, v_{DS} is negative when a reverse current OC fault occurs. As a result, it is now possible for v_R to be greater than v_{GS} , and therefore, special attention must be given when determining the power rating of the charging resistor.

The second component that needs to be reconsidered is the sensing diode, D_{sense} . When the diode is not blocking the drain-to-source voltage, the current through the sensing diode is

$$i_{Dsense} = \frac{v_{GS} - v_{f_Dsense} - v_{DS}}{R_{CHG}} \quad (5)$$

As with the resistor, when a reverse current OC fault occurs, v_{DS} is negative, and i_{Dsense} is greater than it is under standard desaturation protection. As a result, special attention must be given to the maximum current flowing through the diode. In addition, the breakdown voltage of the sensing diode must be greater than the possible off-state voltage of the power device. Furthermore, the diode should be a super-fast recovery diode or a Schottky diode to reduce the noise produced when the sensing diode turns off [8].

The third component that needs to be reconsidered is the blanking capacitor, C_{BLK} . The blanking capacitor should be able to handle a voltage of

$$v_{CBLK} = v_{DS} - v_{Dsense} \quad (6)$$

Special attention should be given during switching transitions due to possible overshoot in voltages.

The fourth component that must be considered is the voltage clamper in the OC branch, D_z . When a reverse current OC fault occurs, the current through the load flows through the OC branch, meaning the load current flows through the voltage clamper. Therefore, the voltage clamper must be able to handle a pulse current equal to the rated current of the load. In addition, the voltage clamper must absorb the energy stored in the load. If the load is inductive, the voltage clamper must be able to handle an energy of

$$E_z = \frac{1}{2} L \cdot (I_{max})^2 \quad (7)$$

where L is the inductance of the load and I_{max} is the maximum instantaneous current flowing through the load.

The fifth component that must be considered is the standard diode in the OC branch, D_{open} . This diode must be sized to handle the voltage of the power device when it is turned off.

TABLE II
DESIGN CONSIDERATIONS

Component	Design Considerations Beyond Traditional Desat
R_{CHG}	Power rating needs to be reconsidered since the voltage across the resistor can be greater than before due to a negative drain-to-source voltage caused by the open-circuit branch
D_{sense}	Special attention must be given to the maximum current flowing through the diode since it is larger than before due to the possible negative drain-to-source voltage
C_{BLK}	- Power rating needs to be reconsidered since the voltage across the capacitor can be greater than before due to a negative drain-to-source voltage caused by the open-circuit branch - Must be able to handle voltage overshoots
D_z	- Must be able to handle a pulse current equal to the rated current of the load - Must be able to absorb the energy stored in the load
D_{open}	- Must be sized to handle the voltage of the power device when it is turned off - Must be able to handle a pulse current equal to the rated current of the load

Like the voltage clamper, the diode in the OC branch must also handle a pulse current equal to the rated current of the load.

The previous analysis of all the design considerations necessary is summarized in Table II.

IV. VERIFICATION

In this section, a simulation and experiment are designed to showcase the new scheme's performance in a two-level, single-phase inverter. Then, the results of running the simulation and conducting the experiment are presented and discussed. Furthermore, a cost analysis is performed to showcase the inexpensive nature of the new solution.

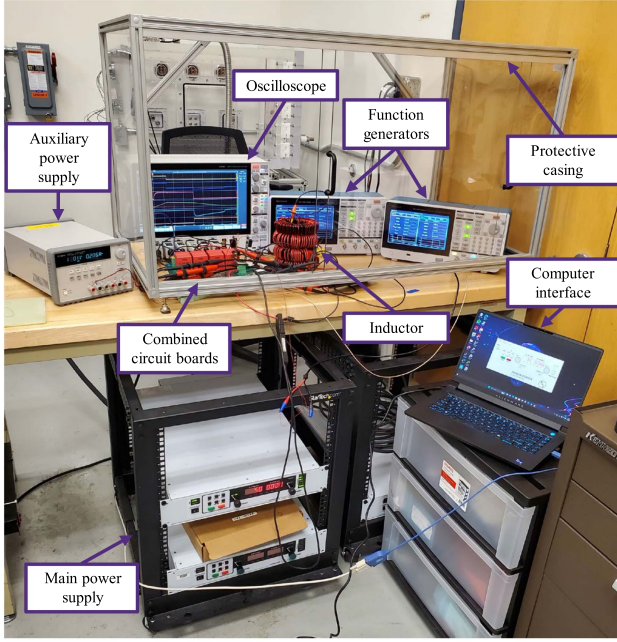
A. Simulated and Experimental Configuration

For comprehensive simulation-related studies under all fault conditions, consult [22] and [23]. However, since the standard desaturation protection scheme detects and protects against SC and forward current OC faults, the simulated and experimental work presented in this paper pertains to reverse current OC faults.

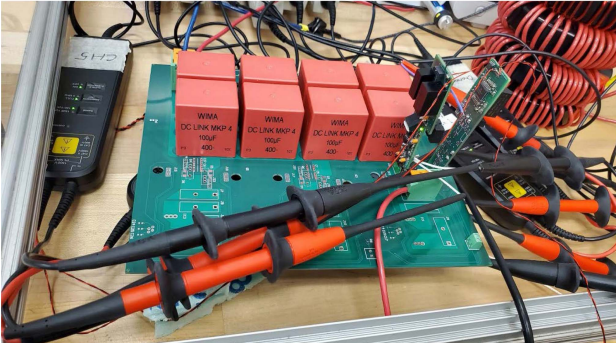
To show the functionality of the newly proposed design, it is simulated and experimentally tested in a two-level, single-phase inverter. The simulation is conducted using Synopsys/Saber software, and the experimental setup makes use of three printed circuit boards (PCBs): the power stage board, the OC fault generation (OCFG) board, and the gate drive board.

The power stage board is the foundational board on which the other two PCBs are built. It includes the sensing diode and the OC fault detection branch.

A reverse current OC fault must be created to showcase the new protection and detection scheme's effectiveness. One way to achieve this is by using an OCFG board. This board allows for a controlled way to open a switch. The switch used in this application is an IGBT without a corresponding freewheeling diode. Hence, when it is opened, there is no alternative path current can flow, emulating an actual OC fault.



(a)



(b)

Fig. 5. Experimental setup showcasing (a) combined experimental circuit and (b) power stage board.

In its most basic form, the gate drive board provides the power to turn the circuit's upper and lower power devices on and off. Additionally, the blanking capacitor, noise immunity diode, noise immunity resistor, charging resistor, and bipolar comparators are all contained on the gate drive board. In this experimental setup, two gate drive boards are required: one for the upper power device and one for the lower power device.

Auxiliary equipment required for experimentation is also necessary. Fig. 5(a) shows the entire circuitry used for experimentation, and Fig. 5(b) shows the constructed power stage board. Fig. 6 gives an equivalent schematic of the circuit tested in the simulation and the experiment.

B. Component Design

All parameter and component information for the critical pieces of the experiment is given in Table III. The values of R_{CHG} , C_{BLK} , and D_{sense} are calculated using the equations in [8], and their ratings are determined by using (3)–(6).

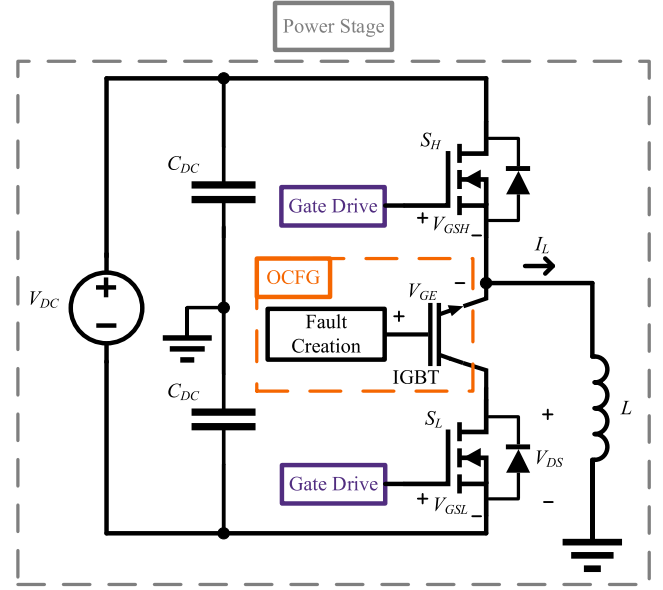


Fig. 6. Equivalent schematic of tested circuit.

TABLE III
PARAMETERS AND COMPONENTS FOR EXPERIMENTAL SETUP

Parameter or Component	Value or Part Number
R_{CHG}	240 Ω , 1/8 W
C_{BLK}	6 nF, 25 V
R_n	240 Ω , 1/8 W
V_{CSTH+}	11.19 V
V_{CSTH-}	-7.90 V
$V_{z,b}$	9 V to 15 V
D_n	Schottky diode, BAT54HT1G
D_{sense}	3.3 kV, GAP3SLT33-214
D_z	200 A surge TVS, SMCJ9.0CA-13-F
D_{open}	3.3 kV, GAP3SLT33-214
L	262 μ H
I_L	5 A
IGBT	1200 V, 50 A, IGW25N120H3FKSA1
MOSFET	1200 V, 100 A, C3M0021120K

Traditionally, C_{BLK} is in the range of 100's of pF. However, in the configuration shown in Fig. 3, the gate drive output flows through R_{CHG} under a SC fault condition. Hence, R_{CHG} is sized to be smaller than traditionally done, and C_{BLK} is sized to be larger than traditionally done. This technique has the advantage of still having an equivalent blanking time of traditional desaturation protection circuits while having a larger C_{BLK} , and hence, the circuit's dv/dt noise immunity performance is improved.

The positive threshold voltage, V_{CSTH+} , is calculated based on the I-V curves of the C3M0021120K power device and the equations in [8]. The components of the OC branch, D_z and D_{open} , are chosen based on their capability to handle the load current, the DC-bus voltage, and the energy stored in the inductor given by (7). Once these components are determined, the OC voltage, V_{OC} , and subsequently, the negative threshold voltage, V_{CSTH-} , are calculated using (1) and (2), respectively.

Finally, the components of the noise immunity branch, R_n and D_n , are selected such that they have minimal impact during normal operation (D_n being a Schottky diode) and provide an

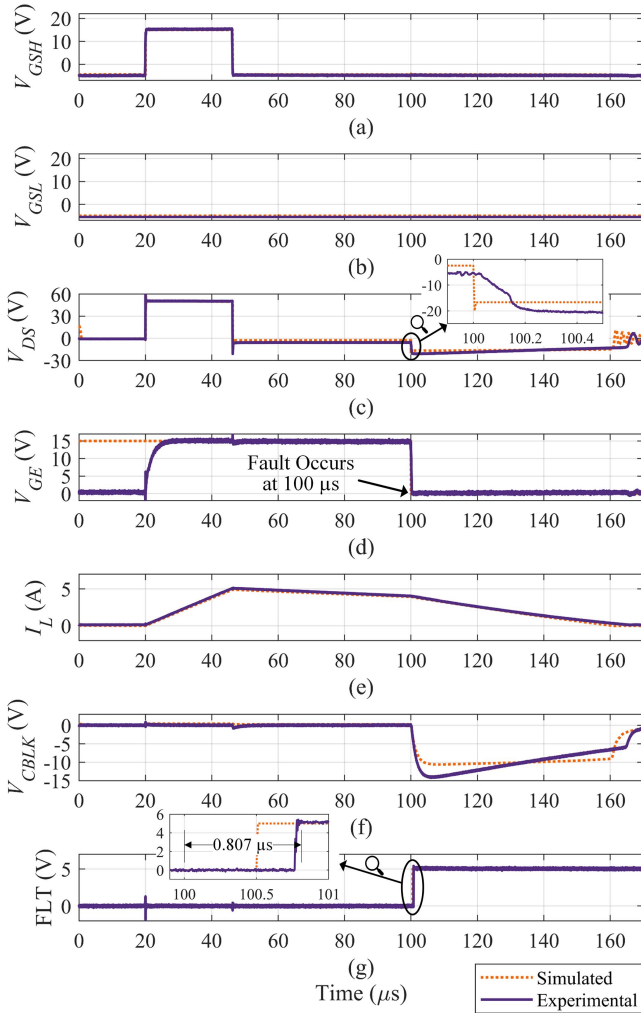


Fig. 7. Simulated versus experimental measurements of (a) upper power device gate-to-source voltage, (b) lower power device gate-to-source voltage, (c) lower power device drain-to-source voltage, (d) IGBT gate-to-emitter voltage, (e) inductor current, (f) blanking capacitor voltage, and (g) fault indicator voltage.

adequate amount of blanking time when a reverse current OC fault occurs (R_n being the same as R_{CHG}). While R_n is typically set to be at least 1 k Ω , by reducing R_n , the blanking time of the circuit when an OC fault occurs is reduced, allowing for faster OC fault detection. The disadvantage of this approach is the slight penalty of a higher current during steady-state operation. Nevertheless, this current is minimal (in the range of mA) due to the bypass diode, and it is unlikely to make a significant difference in overall operation.

C. Results

The results of both tests are shown in Fig. 7. Control signals generated by a function generator are used to control the upper power device and IGBT. These signals produce current through the inductor and create the reverse current OC fault. Due to these signals, the gate-to-source voltage of the upper power device is high (meaning the upper power device is on) from 20 μ s to 46.2 μ s, and the gate-to-emitter voltage of the IGBT

is high (meaning the IGBT is on) until 100 μ s. In response to the gate-to-source voltage of the upper power device, the drain-to-source voltage of the lower power device increases from 0 V to 50 V at 20 μ s and decreases from 50 V back to 0 V at 46.2 μ s. Furthermore, the current flowing through the inductor ramps from 0 A to 5 A from 20 μ s to 46.2 μ s. It then slowly declines from 46.2 μ s to 100 μ s due to the inevitable dissipative elements in the circuit. At 100 μ s, the reverse current OC fault occurs by turning the IGBT off through the OCFG circuitry. At this time, the drain-to-source voltage is clamped to the voltage across the OC branch. Thus, the fault is protected almost instantaneously (no extreme voltage spike occurs). The voltage across the blanking capacitor smoothly approaches this clamped voltage until the negative threshold voltage of -7.90 V is reached. This threshold is reached after 0.51 μ s in the simulation and 0.807 μ s in the experiment, and the fault indicator is consequently set high. During this period, the current through the inductor decreases even faster due to the greater resistance contributed by the OC branch diode, D_{open} . At around 170 μ s, the inductor current reaches 0 A, and the drain-to-source voltage of the lower power device goes to 0 V following some parasitic ringing. This ringing is at a frequency bypassed by the blanking capacitor, so the voltage across the capacitor transitions to 0 V.

This test is successfully repeated over ten times in succession. Therefore, the newly proposed scheme successfully protects against and detects reverse current OC faults.

While most simulated and experimental signals are nearly identical, there are a few minor differences. The voltage across the blanking capacitor in the simulation is initially less than the voltage across the blanking capacitor found in the experiment. This is because a Zener diode is used in simulation, whereas a transient voltage suppressor (TVS) is used in the experiment. Therefore, the voltage across the Zener diode, and consequently the voltage across the blanking capacitor, is more constant (around -10 V) in the simulation. In contrast, the voltage across the TVS, and consequently the voltage across the blanking capacitor, is a function of load current in the experiment. However, the general smoothing effect on the blanking capacitor voltage is the same in the simulation and the experiment due to the circuit's charging resistor and noise immunity section. The fault is detected faster in simulation (0.51 μ s) than it is experimentally (0.807 μ s), but it is protected instantaneously (within 10 ns) in both cases.

A final note about the temperature-independent nature of the solution proposed in this paper is given here. The main impact temperature will have on the novel OC fault protection is a shift in the breakdown voltage of the voltage clamper. However, by adjusting x when computing the negative threshold voltage in (2), the temperature dependence of the breakdown voltage can be circumnavigated. Therefore, the new scheme is robust under the presence of various temperatures.

D. Cost Analysis

Five new components per power device are introduced into the overall circuit to allow for OC fault protection and detection.

Most solutions that require current or voltage sensors do not use pre-existing sensors that are already embedded in converter circuitry. These pre-existing sensors either do not exist or do not have a fast enough sampling rate. Hence, solutions that require a current sensor with a comparable sensing range (like those given in [9] and [10]) and solutions that require a voltage sensor with a comparable sensing range (like that given in [14]) can cost an order of magnitude more than the new scheme as of early 2024. However, this cost difference highly depends on how many power devices are contained in the converter and how many current or voltage sensors are required. The fewer power devices in the converter, the greater the cost-benefit of the solution proposed in this paper.

V. CONCLUSION

This paper proposes a new method for protecting against and detecting all common types of faults in power converter circuitry: short-circuit faults, forward current open-circuit faults, and reverse current open-circuit faults. This is done by integrating an open-circuit fault protection branch, a noise immunity branch, and a bipolar comparator with pre-existing desaturation protection circuitry. A simulation and experimental platform are developed to test the newly proposed design. Experimental results confirm what is found in the simulation. Reverse current open-circuit faults are shown to be protected in a few nanoseconds while being detected in $0.807 \mu\text{s}$. In addition to the fast protection and detection speeds the new scheme produces, it is also cost-efficient due to a complete lack of sensors—only cheaper, easier to obtain components are required. Further, it is universal due to its power device level implementation. Theoretically, it can be embedded in any converter circuitry. Moreover, due to the flexibility of the scheme when designing threshold voltages, it can be viewed as temperature independent. In the future, various types of power devices and topologies can be tested to demonstrate the universal nature of the newly proposed scheme.

REFERENCES

- [1] C. Wan, M. Huang, C. K. Tse, and X. Ruan, "Effects of interaction of power converters coupled via power grid: A design-oriented study," *IEEE Trans. Power Electron.*, vol. 30, no. 7, pp. 3589–3600, Jul. 2015, doi: [10.1109/TPEL.2014.2349936](#).
- [2] C. Wan, M. Huang, C. K. Tse, S.-C. Wong, and X. Ruan, "Nonlinear behavior and instability in a three-phase boost rectifier connected to a nonideal power grid with an interacting load," *IEEE Trans. Power Electron.*, vol. 28, no. 7, pp. 3255–3265, Jul. 2013, doi: [10.1109/TPEL.2012.2227505](#).
- [3] P. America, "About wide bandgap semiconductors," Power America Institute, 2018. Accessed: Jul. 2021. [Online]. Available: <https://poweramericainstitute.org/about-poweramerica/lets-learn-about-power-electronics/>
- [4] I. Kizilyalli, Z. Shen, and D. Cunningham, *Direct Current Fault Protection: Basic Concepts and Technology Advances*. Berlin, Germany: Springer-Verlag, 2023.
- [5] P. Hacke et al., "A status review of photovoltaic power conversion equipment reliability, safety, and quality assurance protocols," *Renewable Sustain. Energy Rev.*, vol. 82, pp. 1097–1112, Feb. 2018, doi: [10.1016/j.rser.2017.07.043](#).
- [6] Z. Wang et al., "Temperature-dependent short-circuit capability of silicon carbide power MOSFETs," *IEEE Trans. Power Electron.*, vol. 31, no. 2, pp. 1555–1566, Feb. 2016, doi: [10.1109/TPEL.2015.2416358](#).
- [7] H. Li et al., "E-mode GaN HEMT short circuit robustness and degradation," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2017, pp. 1995–2002, doi: [10.1109/ECCE.2017.8096401](#).
- [8] F. Wang and Z. Zhang, "Protection design for double pulse test," in *Characterization of Wide Bandgap Power Semiconductor Devices*, vol. 128, (IET Energy Engineering Series), London, U.K.: Inst. of Energy and Technol., 2018, pp. 141–163.
- [9] T. Kamel, Y. Biletskiy, and L. Chang, "Fault diagnoses for industrial grid-connected converters in the power distribution systems," *IEEE Trans. Ind. Electron.*, vol. 62, no. 10, pp. 6496–6507, Oct. 2015, doi: [10.1109/TIE.2015.2420627](#).
- [10] U. M. Choi, J.-S. Lee, F. Blaabjerg, and K.-B. Lee, "Open-circuit fault diagnosis and fault-tolerant control for a grid-connected NPC inverter," *IEEE Trans. Power Electron.*, vol. 31, no. 10, pp. 7234–7247, Oct. 2016, doi: [10.1109/TPEL.2015.2510224](#).
- [11] A. Anand, N. Raj, S. George, and J. G., "Open switch fault detection in cascaded H-bridge Multilevel Inverter using normalised mean voltages," in *Proc. IEEE 6th Int. Conf. Power Syst.*, 2016, pp. 1–6, doi: [10.1109/ICPES.2016.7584128](#).
- [12] J. He, N. A. O. Demerdash, N. Weise, and R. Katebi, "A fast on-line diagnostic method for open-circuit switch faults in SiC-MOSFET-based T-type multilevel inverters," *IEEE Trans. Ind. Appl.*, vol. 53, no. 3, pp. 2948–2958, May/Jun. 2017, doi: [10.1109/TIA.2016.2647720](#).
- [13] G. Guruvareddiyar and B. Subathra, "An open and short circuit switch fault of power converters in PM-BLDC motor in electric vehicle," in *Proc. IEEE Int. Conf. Clean Energy Energy Efficient Electron. Circuit Sustain. Develop.*, 2019, pp. 1–3, doi: [10.1109/INCCES47820.2019.9167737](#).
- [14] S. Yang, Y. Tang, and P. Wang, "Seamless fault-tolerant operation of a modular multilevel converter with switch open-circuit fault diagnosis in a distributed control architecture," *IEEE Trans. Power Electron.*, vol. 33, no. 8, pp. 7058–7070, Aug. 2018, doi: [10.1109/TPEL.2017.2756849](#).
- [15] Y. Xia, Y. Xu, and B. Gou, "A data-driven method for IGBT open-circuit fault diagnosis based on hybrid ensemble learning and sliding-window classification," *IEEE Trans. Ind. Informat.*, vol. 16, no. 8, pp. 5223–5233, Aug. 2020, doi: [10.1109/TII.2019.2949344](#).
- [16] Z. Li, Y. Gao, X. Zhang, B. Wang, and H. Ma, "A model-data-hybrid-driven diagnosis method for open-switch faults in power converters," *IEEE Trans. Power Electron.*, vol. 36, no. 5, pp. 4965–4970, May 2021, doi: [10.1109/TPEL.2020.3026176](#).
- [17] J. Lamb and B. Mirafzal, "Open-circuit IGBT fault detection and location isolation for cascaded multilevel converters," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 4846–4856, Jun. 2017, doi: [10.1109/TIE.2017.2674629](#).
- [18] R. W. Erickson and D. Maksimovic, *Fundamentals of Power Electronics*. Berlin, Germany: Springer-Verlag, 2001.
- [19] J. P. Colinge and C. A. Colinge, *Physics of Semiconductor Devices*. Norwell, MA, USA: Kluwer, 2002.
- [20] R. Anderson and B. Anderson, *Fundamentals of Semiconductor Devices*. New York, NY, USA: McGraw Hill, 2004.
- [21] X. Huang et al., "Comprehensive analysis and improvement methods of noise immunity of desat protection for high voltage SiC MOSFETs with high DV/DT," *IEEE Open J. Power Electron.*, vol. 3, pp. 36–50, 2022, doi: [10.1109/OJPEL.2021.3134498](#).
- [22] B. Brown, "Universal short-circuit and open-circuit fault detection for an inverter," M.S. thesis, College Elect. Comput. Eng., Clemson Univ., Clemson, SC, USA, 2023.
- [23] B. Brown and Z. Zhang, "Universal short-circuit and open-circuit fault detection for an inverter," in *Proc. 11th Int. Conf. Power Electron. ECCE Asia*, 2023, pp. 3230–3238, doi: [10.23919/ICPE2023-ECCEAsia54778.2023.10213883](#).



Buck F. Brown III (Graduate Student Member, IEEE) received the B.S. and M.S. degrees in electrical engineering in 2021 and 2023, respectively, from Clemson University, Clemson, SC, USA, where he is currently working toward the Ph.D. degree in electrical engineering. He is a Research Assistant of power electronics with Clemson University. His research interests include the use of control systems in power electronics, renewable energy, inverters, and the study and use of advanced electronics for power systems control. His honors and awards include the Palmetto Fellow's and Charles C. Kanapaux Electrical Engineering Scholarships. He was the President of Eta Kappa Nu and is also a member of Tau Beta Pi.



Zheyu Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the Huazhong University of Science and Technology, Wuhan, China, and the Ph.D. degree in electrical engineering from the University of Tennessee, Knoxville, TN, USA, in 2008, 2011, and 2015, respectively. He is currently an Assistant Professor with Rensselaer Polytechnic Institute, Troy, NY, USA. From 2018 to 2019, he joined General Electric Research as the Lead Power Electronics Engineer with Niskayuna, NY. Afterward, he was the Warren H. Owen-Duke

Energy Assistant Professor of engineering with Clemson University, Clemson, SC, USA, from 2019 to 2023. Dr. Zhang is also an Associate Editor for the IEEE TRANSACTIONS ON POWER ELECTRONICS and IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS. He was the recipient of two prize paper awards from the IEEE Industry Applications Society and IEEE Power Electronics Society.



Johan H. Enslin (Life Fellow, IEEE) received the Ph.D. degree in electrical and electronic engineering from Rand Afrikaans University (RAU), Johannesburg, South Africa, in 1988. He is currently the Duke Energy Endowed Chaired Professor of smart grid with Clemson University, North Charleston, SC, USA, and an Executive Director for the Energy Systems Program with Zucker Family Graduate Education Center, North Charleston. Dr. Enslin is a Fellow of the South African Institute of Electrical Engineers. He is the VP Standards for the IEEE Power Electronics Society. He

is a registered Professional Engineer in South Africa.



Cheng Wan (Senior Member, IEEE) received the B.S. degree in hydraulic and hydropower engineering from Wuhan University, Wuhan, China, in 2009, and the Ph.D. degree in electrical and electronic engineering from the Huazhong University of Science and Technology, Wuhan, in 2014. He is currently a Research Scientist with Clemson University, Charleston, SC, USA. He previously held the position of Postdoctoral Fellow with Clemson University. Before joining Clemson University, he was the power electronics senior Specialist with APTIV Company, Ltd. With

more than ten years of international company R&D experience, he began his career with United Imaging Healthcare Ltd., Shanghai, China, and subsequently worked for LG Electronics, Huawei, Eaton, and APTIV Company, Ltd. His research interests include power electronics, renewable energy generation and stability, electrical vehicle power conversion, and motor drive. Dr. Wan was the recipient of a Prize Paper Award from the IEEE Power Electronics Society, for his contributions.